SINGAPORE DIMENSIONS • KINDERGARTEN (KA) • CHAPTER 4

Number Bonds to 5 & 10

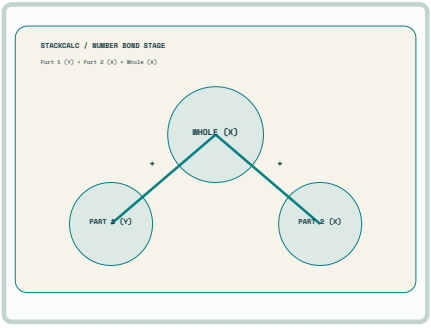
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.K.OA.A.3

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Number Bond Stage

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place a 3|2 domino in the tray dock. Left half is 3, right half is 2.

Sketch Model on Paper:
Draw a number bond circle with Whole connected to two Part circles.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	4	X: 4	Key 1st operand
2.	ENTER	Y: 4 | X: 4	Stack lift copies X
3.	3	Y: 4 | X: 3	Key 2nd operand
4.	+	X: Whole	Drop stack & compute

EXPECTED LCD X: Whole is 7

1 **Number bond: Part 1 is 5, Part 2 is 2. Whole?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Number bond: Part 1 is 6, Part 2 is 4. Whole?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Whole is 9, Part 1 is 5. Find missing Part 2.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Whole is 10, Part 1 is 3. Find missing Part 2.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 4

Number Bonds to 5 & 10 • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Math Domino Studio (12-Frame Tray) + Dominoes

CAD Blueprint: number_bond_stage

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/math_dominoes_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal & Warm White (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Number bond: Part 1 is 5, Part 2 is 2. Whole** Standard: CCSS.MATH.K.OA.A.3	7	[5] [ENTER] [2] [+]
2	**Number bond: Part 1 is 6, Part 2 is 4. Whole** Standard: CCSS.MATH.K.OA.A.3	10	[6] [ENTER] [4] [+]
3	**Whole is 9, Part 1 is 5. Find missing Part 2** Standard: CCSS.MATH.K.OA.A.3	4	[9] [ENTER] [5] [-]
4	**Whole is 10, Part 1 is 3. Find missing Part** Standard: CCSS.MATH.K.OA.A.3	7	[1][0] [ENTER] [3] [-]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

To find a missing part, emphasize entering Whole first, then Part, then [-].

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.